

Machine Learning (CAI-262)

Objectives:

- Discriminant based classification.
- Support Vector Machines
- Finding Margin of a Linear Classifier
- SVM as Optimization problem
- Lagrange variables

Discriminant based classification:

In this type of classification, the objective is to learn a function or, in the case of more than 2 classes, a set of functions from training data which can generate decisions for test data such that the classes in the data can be separated.

SVM use a linear discriminant function to differentiate between patterns of different classes that is why SVM is a linear classifier.

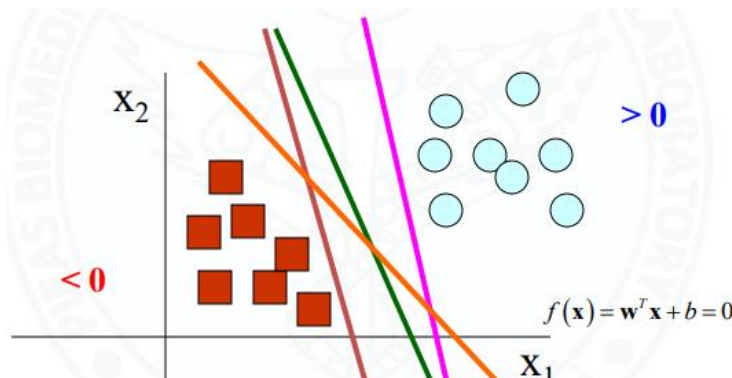
Support Vector Machine (SVM):

Support Vector Machine or SVM is one of the most popular Supervised Learning algorithms, which is used for Classification as well as Regression problems.

The goal of the SVM algorithm is to create the best line or decision boundary that can segregate n-dimensional space into classes so that we can easily put the new data point in the correct category in the future. This best decision boundary is called a **hyperplane**.

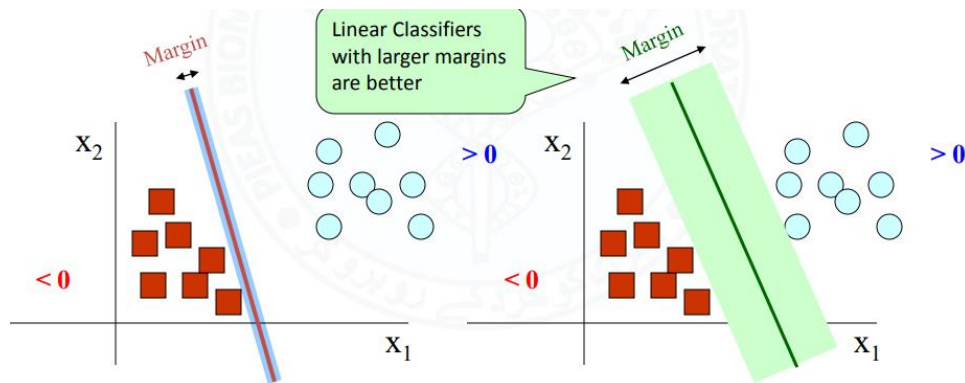
SVM chooses the extreme points/vectors that help in creating the hyperplane. These extreme cases are called as **support vectors**, and hence algorithm is termed as Support Vector Machine. Consider the below diagram in which there are two different categories that are classified using a decision boundary or hyperplane:

There are a large number of lines (or in general 'hyperplanes') separating the two classes.



The boundary which lies at the maximum distance from data points of both classes gives better tolerance to noise and better generalization.

The width by which the boundary of a linear classifier can be increased before hitting a data point is called the **margin of the linear classifier**.



Support Vector Machines are **linear classifiers** that produce the optimal separating boundary (hyper-plane). It tries to find $\mathbf{w}$ and $\mathbf{b}$ in a way so as to maximize the margin while classifying all the training patterns correctly (for linearly separable problem).

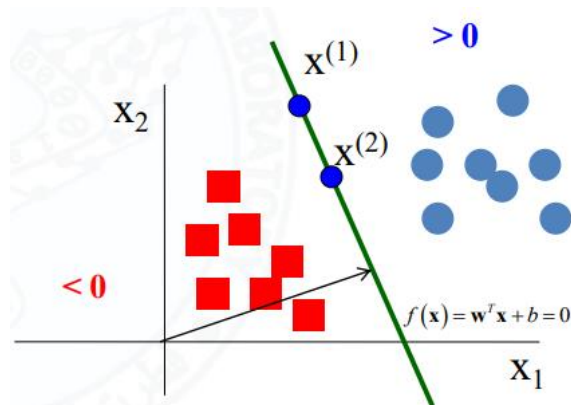
Finding Margin of a Linear Classifier:

Consider a linear classifier with the boundary.

$$f(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + b = 0 \quad \text{for all } \mathbf{x} \text{ on the boundary}$$

We know that the vector $\mathbf{w}$ is perpendicular to the boundary.

Consider two points $\mathbf{x}^{(1)}$ and $\mathbf{x}^{(2)}$ on the boundary.



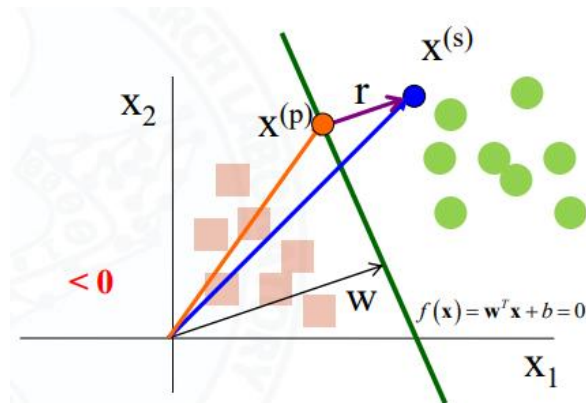
$$f(\mathbf{x}^{(1)}) = \mathbf{w}^T \mathbf{x}^{(1)} + b = 0 \quad (1)$$

$$f(\mathbf{x}^{(2)}) = \mathbf{w}^T \mathbf{x}^{(2)} + b = 0 \quad (2)$$

Subtracting (1) from (2)

$$\mathbf{w}^T (\mathbf{x}^{(2)} - \mathbf{x}^{(1)}) = 0 \quad \Rightarrow \quad \mathbf{w} \perp (\mathbf{x}^{(2)} - \mathbf{x}^{(1)})$$

Let $\mathbf{x}^{(s)}$ be a point in the feature space with its projection $\mathbf{x}^{(p)}$ on the boundary.



$$f(\mathbf{x}^{(p)}) = \mathbf{w}^T \mathbf{x}^{(p)} + b = 0$$

We know that

$$\begin{aligned} \mathbf{x}^{(s)} &= \mathbf{x}^{(p)} + r\hat{\mathbf{w}} \\ \Rightarrow f(\mathbf{x}^{(s)}) &= \mathbf{w}^T \mathbf{x}^{(s)} + b \\ &= \mathbf{w}^T (\mathbf{x}^{(p)} + r\hat{\mathbf{w}}) + b \\ &= \mathbf{w}^T \mathbf{x}^{(p)} + r\mathbf{w}^T \hat{\mathbf{w}} + b \\ &= \mathbf{w}^T \mathbf{x}^{(p)} + b + r\mathbf{w}^T \frac{\mathbf{w}}{\|\mathbf{w}\|} \\ &= 0 + r\|\mathbf{w}\| = r\|\mathbf{w}\| \end{aligned}$$

$$\Rightarrow r = \frac{f(\mathbf{x}^{(s)})}{\|\mathbf{w}\|}$$

The points in the training set that lie closest (having minimum perpendicular distance) to the separating hyper-plane are called **support vectors**.

Margin is twice of perpendicular distances of the hyper-plane from the nearest support vector of the two classes.

$$\rho = 2|r| = 2 \left| \frac{f(\mathbf{x}^*)}{\|\mathbf{w}\|} \right|$$

Nearest Support Vector

Assume that all data is at least distance 1 from the hyperplane, then the following two constraints follow for a training set $\{(\mathbf{x}^{(i)}, y_i)\}$

$$\mathbf{w}^T \mathbf{x}^{(i)} + b \geq 1 \quad \text{if } y_i = 1$$

$$\mathbf{w}^T \mathbf{x}^{(i)} + b \leq -1 \quad \text{if } y_i = -1$$

$$\Rightarrow \rho = 2|r| = 2 \left| \frac{f(\mathbf{x}^*)}{\|\mathbf{w}\|} \right| = 2 \left| \frac{\pm 1}{\|\mathbf{w}\|} \right|$$

$$\rho = \frac{2}{\|\mathbf{w}\|}$$

We can present SVM as the following optimization problem.

$$\begin{aligned} \max \quad & \rho = \frac{2}{\|\mathbf{w}\|} \\ \text{s.t.} \quad & \mathbf{w}^T \mathbf{x}^{(i)} + b \leq -1, \quad \forall i \text{ s.t. } y_i = -1 \\ & \mathbf{w}^T \mathbf{x}^{(i)} + b \geq +1, \quad \forall i \text{ s.t. } y_i = +1 \end{aligned}$$

OR

$$\begin{aligned} \min \quad & \rho = \frac{1}{2} \mathbf{w}^T \mathbf{w} \\ \text{s.t.} \quad & y_i (\mathbf{w}^T \mathbf{x}^{(i)} + b) \geq +1 \end{aligned}$$

SVM as Optimization problem:

In the process of maximizing the margin, we need to solve a constrained optimization problem. For that purpose, we use the method of Lagrange multipliers. So first take a look at that.

Method of Lagrange multipliers for constrained optimization:

Given a cost function $J(\mathbf{x}, \mathbf{y})$, suppose that we are trying to find the minimum of that function with respect to the constraint $g(\mathbf{x}, \mathbf{y})=0$. Using the method of Lagrange multipliers:

1. Define a new function
 $L(\mathbf{x}, \mathbf{y}, \lambda) = J(\mathbf{x}, \mathbf{y}) - \lambda(g(\mathbf{x}, \mathbf{y}) - c)$, where $\lambda \geq 0$
2. Find the partial derivatives and equate them to zero
 $\partial L / \partial \mathbf{x} = 0$, and $\partial L / \partial \mathbf{y} = 0$
3. Drop out λ by using these two equations to find relations between $\mathbf{x}$ and $\mathbf{y}$.
4. Substitute this relation into the constraint and solve the equation for $\mathbf{x}$ and $\mathbf{y}$

Example:

Find the minimum value of the function

$$f(x, y) = x^2 + y^2 - 2x - 2y$$

Subject to the constraint

$$x + 2y = 4$$

Solution:

$$L(x, y, \lambda) = x^2 + y^2 - 2x - 2y - \lambda(x + 2y - 4)$$

$$\frac{\partial L}{\partial x} = 2x - 2 - \lambda = 0 \implies \lambda = 2x - 2$$

$$\frac{\partial L}{\partial y} = 2y - 2 - \lambda = 0 \implies \lambda = y - 1$$

When we eliminate λ from the equations, we get: $y = 2x - 1$

Substitute y in the constraint: $x + 2(2x - 1) = 4 \implies x = 6/5$

Since $y = 2x - 1 \implies y = 12/5 - 1 = 7/5$

Thus, the value of the function f subject to the constraint $2x + y = 4$ is:

$$f(x, y) \Big|_{\substack{x=6/5 \\ y=7/5}} = (6/5)^2 + (7/5)^2 - 2(6/5) - 2(7/5) = -9/5$$

Support Vector Machine (SVM):

In order to maximize the margin $\frac{2}{\|w\|}$, we will minimize the square of the norm of the weight vector, The cost function is the defined as $\frac{1}{2} \|w\|^2$.

Since we would like to have $w^T x_i + b = y_i$, in order to correctly classify the sample x_i , we can use this expression as a constraint to minimize the cost function.

$$L(w, b, \lambda) = \frac{1}{2} w^T w - \sum_{i=1}^m \lambda_i [y_i (w^T x_i + b) - 1],$$

$$\forall_i \lambda_i \geq 0$$

Let us find the gradients with respect to the weight vector and bias and equate them to zero.

$$\frac{\partial L}{\partial w} = 0 \implies w - \sum_{i=1}^m \lambda_i y_i x_i = 0 \implies w = \sum_{i=1}^m \lambda_i y_i x_i$$

$$\frac{\partial L}{\partial b} = 0 \implies \sum_{i=1}^m \lambda_i y_i = 0$$

The above formulation is called the Lagrangian formulation for SVM.

When we expand the terms the minimization problem becomes

$$\min(L_P(\mathbf{w}, b, \lambda)) = \frac{1}{2} \mathbf{w}^T \mathbf{w} - \sum_{i=1}^m \lambda_i y_i \mathbf{w}^T \mathbf{x}_i - \sum_{i=1}^m \lambda_i y_i b + \sum_{i=1}^m \lambda_i$$

subject to $\forall_i \lambda_i \geq 0$

When we insert the expressions into above equation and maximize with respect to λ values we obtain the dual problem.

$$\max(L_D(\lambda_i)) = \frac{1}{2} \sum_{i=1}^m \lambda_i y_i \lambda_j y_j (\mathbf{x}_i^T \mathbf{x}_j) - \sum_{i=1}^m \lambda_i y_i \lambda_j y_j (\mathbf{x}_i^T \mathbf{x}_j) - (0)b + \sum_{i=1}^m \lambda_i,$$

When we simplify the dual formulation we get.

$$\max(L_D(\lambda_i)) = \sum_{i=1}^m \lambda_i - \frac{1}{2} \sum_{i=1}^m \lambda_i y_i \lambda_j y_j (\mathbf{x}_i^T \mathbf{x}_j)$$

subject to $\sum_{i=1}^m \lambda_i y_i = 0$ & $\lambda_i \geq 0$

Note that the dual formulation can be solved using quadratic optimization (such as gradient ascent) by taking the derivatives of L with respect to every λ . In this formulation we only need to compute the dot product of the feature vector $\mathbf{x}_i$.

Example:

Consider a data set with a positive class as follows:

$$\left\{ \begin{pmatrix} 3 \\ 1 \end{pmatrix}, \begin{pmatrix} 3 \\ -1 \end{pmatrix}, \begin{pmatrix} 6 \\ 1 \end{pmatrix}, \begin{pmatrix} 6 \\ -1 \end{pmatrix} \right\}$$

Consider a data set with the negative class as follows:

$$\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \end{pmatrix} \right\}$$

The support is given as follows:

$$s_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, s_2 = \begin{pmatrix} 3 \\ 1 \end{pmatrix}, s_3 = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$$

As a first step append a bias of 1 with each vector:

$$\tilde{s}_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \tilde{s}_2 = \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix}, \tilde{s}_3 = \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix}$$

Applying the SVM equation would result in:

$$\alpha_1 \tilde{s}_1 \tilde{s}_1 + \alpha_2 \tilde{s}_1 \tilde{s}_2 + \alpha_3 \tilde{s}_1 \tilde{s}_3 = -1$$

$$\alpha_1 \tilde{s}_1 \tilde{s}_2 + \alpha_2 \tilde{s}_2 \tilde{s}_2 + \alpha_3 \tilde{s}_2 \tilde{s}_3 = 1$$

$$\alpha_1 \tilde{s}_1 \tilde{s}_3 + \alpha_2 \tilde{s}_2 \tilde{s}_3 + \alpha_3 \tilde{s}_3 \tilde{s}_3 = 1$$

It could be noted that the first point is on the negative class while the next two are positive therefore first one is negative and the other two are positive.

$$\alpha_1 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + \alpha_2 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix} + \alpha_3 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix} = -1$$

$$\alpha_1 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix} + \alpha_2 \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix} \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix} + \alpha_3 \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix} \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix} = 1$$

$$\alpha_1 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix} + \alpha_2 \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix} \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix} + \alpha_3 \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix} \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix} = 1$$

Taking dot products of the vectors would result in:

$$\begin{aligned} 2\alpha_1 + 4\alpha_2 + 4\alpha_3 &= -1 \\ 4\alpha_1 + 11\alpha_2 + 9\alpha_3 &= +1 \\ 4\alpha_1 + 9\alpha_2 + 11\alpha_3 &= +1 \end{aligned}$$

The above three equations have three unknowns that could be solved using elimination, substitution, or Gaussian elimination. The resulting values of α are, $\alpha_1 = -3.5$, $\alpha_2 = 0.75$ and $\alpha_3 = 0.75$. To find the decision boundary we need to compute the weights which are given by the equation

6.17.

$$\tilde{w} = \sum_i \alpha_i \tilde{s}_i$$

$$\Rightarrow -3.5 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + 0.75 \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix} + 0.75 \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}$$

The equation of straight line $y = mx + c$ has $m = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $c = -2$.